Multiplying a polynomial by a monom

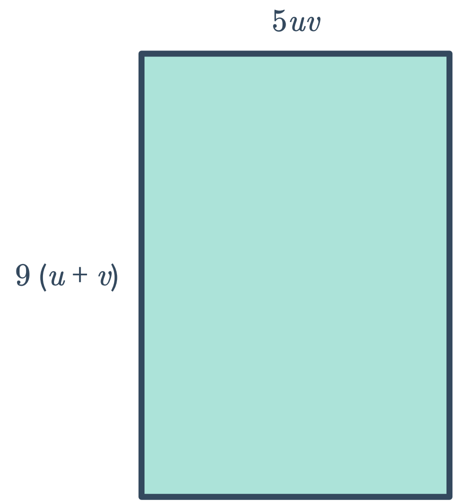
1 Simplify the following expressions:

a $r\left(r+5\right)$	**b** $w\left(1+w\right)$
c $y\left(y-9\right)$	**d** $-m\left(m+1\right)$
e $t\left(2t-14\right)$	**f** $m\left(1-m\right)$
g $2n\left(n-14\right)$	**h** $6y\left(5y-10\right)$
i $-5t\left(3t+8\right)$	**j** $2.3y\left(1.9y+4\right)$
k $-12m\left(\frac{m}{6}+4\right)$	**l** $\frac{y}{4}\left(4y+20\right)$
m $7y\left(y^2+8\right)$	**n** $\left(3y-2\right)\left(-5y^2\right)$
o $10u^2\left(7u^4+8u^3\right)$	**p** $6u^7\left(9u^7+9u^6\right)$
q $7wy\left(y+w\right)$	**r** $8a\left(a-4b-5c\right)$
s $\left(6a+7b-4\right)\left(2a\right)$	**t** $2a\left(5a^2+2a+3\right)$
u $\left(a^2b^2+4ab-6\right)\left(-4ab\right)$	**v** $\left(2a^3+5a^2+2\right)\left(-5ab^2\right)$

2 Simplify the following expressions:

a $y\left(y-4\right)+10$	**b** $2y\left(9y+10y\right)$
c $8u\left(10u+v\right)-5$	**d** $x\left(x+7\right)+3\left(x+10\right)$
e $x\left(x+6\right)+4\left(5x+3\right)$	**f** $9y\left(y+6\right)+8y-2$
g $u\left(u+2\right)-9\left(u+8\right)$	**h** $5+8y\left(y+6\right)-4y^2$
i $2y^2+4y+5y\left(y-7\right)$	**j** $-6u\left(8u-3v\right)+6uv$
k $7y-8y\left(y-2\right)+6y^2$	**l** $6x\left(4x+4\right)+x\left(x+4\right)$
m $\left(6y+4\right)5y+4y\left(y+6\right)$	**n** $4a\left(2a+b\right)+a\left(3a+4b\right)$
o $-3p\left(-10p+2q\right)-p^2-4pq$	**p** $3u\left(3u+3v\right)-u\left(2u-v\right)$
q $5wy\left(y+w\right)+4wy\left(y+2\right)$	**r** $9x\left(4x+8y\right)-2x\left(4y-4x\right)$
s $-5y^2\left(y-5\right)-8y^2+3$	**t** $3y+6y^3-7y\left(y^2-5\right)$
u $3n^2\left(n-2\right)-2n\left(n^2+8\right)$	**v** $4x\left(5x\left(x-3\right)+2x\right)$
w $4z\left(4z^2-3z+6\right)-z^2\left(z-8\right)$	**x** $3\left(b^2-\left(12b-6b\left(2+4b\right)\right)\right)$
y $2\left(x^2-1\right)+4x\left(x-3\right)$	**z** $\left(5c^2-2\right)\left(3-2c+2c^2\right)$

3 Consider the rectangle shown.
Write a fully simplified expression for the area of the rectangle.



4 Neil has a rectangular sheet of wrapping paper that is $5pq$ cm wide and $10p$ cm long. He cuts off a sheet that is $5pq$ cm wide and $7q$ cm long to wrap a gift.
Write a fully simplified expression for the area of the wrapping paper that is leftover.

5 Christa's technology retail store sells hard drives for $\$y$ each. She estimates that if she increases the price of each hard drive by $\$5$, she will sell a total of $\frac{y^4}{5}$ hard drives.
Write a fully simplified expression for what her total revenue will be with the new price.

Additional questions

6 Simplify each product:

a $(x-2)(x-3)$

b $(a+9)(b-6)$

c $(7y-6)(6y+6)$

d $9(y+8)(y+2)$

e $(v+5)\left(5v^2-3v-5\right)$

f $(m-5)\left(4m-3+m^2\right)$

7 A rectangular garden has a length that is one foot less than twice the width.

a Write and simplify an expression for the area of the rectangle.

b A 2 foot border is to be placed all around the garden. Write and simplify an expression for the area of the border.

c The landscaper designing the garden has 100 square feet of pavers to use for the border. What is the largest garden the landscaper can border before running out of pavers? Assume the landscaper will only use positive-integer dimensions.